

SKYRUNNER

Product Requirements Document

Phase 1 & Phase 2 MVP Specifications

Version: 1.0 (April 16, 2026)

Owner: Adaobi Chiamaka Onyekaba, Founder & CEO

Review Audience: Obinna / Attarhi Limited

Build Stack: Cursor Pro IDE, Claude Code Max, Next.js, React Native, Node.js, PostgreSQL, AI
Logistics Layer (Phase 2)

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1. Product Vision

1.1 Vision Statement

Skyrunner delivers the future of premium last-mile logistics to Lagos by replacing unreliable dispatch rider networks with an autonomous drone delivery platform serving the most affluent residential zones in West Africa — starting with Banana Island Estate and expanding to Ikoyi and Lekki.

1.2 Mission

Provide Lagos premium residents with delivery that is fast, secure, transparent, and worthy of the lifestyle they expect — powered by an AI-driven autonomous logistics platform that learns, optimizes, and scales efficiently across premium residential markets.

1.3 Success Metrics (North Star)

- Soft launch on Banana Island Estate by June 30, 2026 with 30 operational drones
- 50 active paying customers within first 30 days post-launch
- 95 percent on-time delivery rate during operational hours
- Zero package theft or loss incidents
- Average customer rating 4.5 stars or higher
- NGN 8M to NGN 15M revenue in first quarter post-launch

2. Users and Personas

2.1 Primary Persona: Premium Resident (Customer)

Name: Chidinma Okafor, 38, Banana Island Estate resident

Profile: Executive at a Nigerian financial services firm, mother of two, married to a corporate lawyer. Household income equivalent to USD 180,000+. Values convenience, discretion, reliability, and time savings.

Pain Points

- Spends 2 to 4 hours weekly coordinating household deliveries that fail or arrive damaged
- Does not trust dispatch riders with packages over NGN 50,000 in value
- Frequently sends her driver to collect items that should have been delivered
- Tired of delays caused by traffic and security gate bottlenecks

Success Looks Like

- Order placed in under 2 minutes through mobile app
- Real-time visibility into delivery status
- Package arrives within 20 minutes sealed and secure
- Payment handled seamlessly through trusted Nigerian processors

2.2 Secondary Persona: Operations Administrator

Name: Tunde, 32, Attarhi Limited operations manager

Profile: Manages day-to-day Skyrunner dispatch from Attarhi office. Needs clear operational visibility without technical complexity.

Pain Points

- Cannot manage complex multi-order dispatch manually at scale
- Needs immediate visibility when drones go offline or pilots miss shifts
- Must coordinate with vendor partners without friction

2.3 Tertiary Persona: Drone Pilot

Name: Emeka, 29, certified drone pilot

Profile: Licensed UAS operator with 2+ years flight experience. Works Skyrunner shifts based on published schedule.

Needs

- Simple shift schedule visibility
- Clear flight plan and cargo handoff protocols
- Performance tracking for skill development and compensation

3. Phase 1 Functional Requirements

Phase 1 build window: April 16 — May 28, 2026 (6 weeks). Primary builder: Adaobi using Cursor Pro IDE and Claude Code Max.

3.1 Customer Mobile Application

REQ-CM-001: User Registration and Authentication

- User registers with full name, phone number, and email
- Phone number verified via SMS OTP through Twilio Africa
- User creates secure password (minimum 8 characters, mixed case, number)
- User accepts Terms of Service and Privacy Policy at registration
- Session persistence via JWT tokens, 30-day refresh window

REQ-CM-002: Address Verification (Banana Island Estate Only in Phase 1)

- User enters residential address within Banana Island Estate
- Address validated against Attarhi Estate resident registry
- Non-Estate residents receive message: Skyrunner is currently serving Banana Island Estate residents only. Waitlist sign-up available.
- Verified residents can save up to 3 delivery addresses

REQ-CM-003: Vendor Catalog Browsing

- Catalog displays 3 to 5 initial vendor partners
- Each vendor has branded storefront with product listings, prices in NGN, photos, descriptions
- Search and filter by vendor, category, price range
- Add to cart with quantity selection
- Cart persists across app sessions

REQ-CM-004: Custom Delivery Request

- User can request delivery from any Lagos source outside vendor catalog
- User enters pickup address, item description, estimated value, recipient name
- Platform calculates estimated delivery fee based on distance and value tier
- User confirms and submits custom request for operator review

REQ-CM-005: Order Placement and Payment

- User reviews cart with itemized breakdown, delivery fee, total
- User selects delivery address from saved addresses or adds new
- Payment integrated with Paystack sandbox initially, production by Week 4
- Supported payment methods: card, bank transfer, Paystack wallet
- Order confirmation screen with order ID and estimated delivery time

REQ-CM-006: Order Status and Notifications

- User receives push notification at status changes: confirmed, dispatched, in transit, delivered
- In-app order tracking page with status timeline
- Estimated delivery time updates based on operator dispatch

REQ-CM-007: Order History

- User views all past orders with date, vendor, items, total, status
- Tap into past order to view full details and receipt

3.2 Operator Dashboard (Web Application)

REQ-OD-001: Admin Authentication

- Admin login restricted to designated Skyrunner team members
- Role-based access: Super Admin (Adaobi), Operations (Attarhi designees), Support
- Two-factor authentication required for all admin accounts
- Session timeout after 60 minutes inactivity

REQ-OD-002: Incoming Order Queue

- Real-time order queue sorted by time received
- Each order card displays: customer name, vendor, items, total, delivery address, status
- Operator can accept or flag order for review
- Accepted orders move to dispatch view

REQ-OD-003: Manual Dispatch Assignment

- Operator selects available drone from fleet list
- Operator assigns pilot from on-shift roster
- System generates dispatch ticket with pickup and delivery details
- Dispatch notification sent to assigned pilot via SMS and dashboard alert

REQ-OD-004: Fleet Status View

- List view of all drones with status: available, in flight, charging, maintenance, offline
- Manual status update capability for Phase 1 (automated in Phase 2)
- Battery level indicator per drone (manual entry in Phase 1)

REQ-OD-005: Pilot Shift Calendar

- Weekly calendar view of pilot shifts
- Admin can add, edit, remove shifts
- Pilot receives shift confirmation via SMS

REQ-OD-006: Daily Revenue and Transaction Log

- Dashboard widget displaying today's revenue, order count, average order value
- Transaction log with filtering by date range, vendor, payment status
- Export to CSV for financial reconciliation

3.3 Infrastructure and Backend Requirements

REQ-INF-001: Hosting and Deployment

- Backend deployed on AWS or Vercel with auto-scaling
- Database: PostgreSQL with managed backup and point-in-time recovery
- File storage: AWS S3 or Vercel Blob for vendor product images, receipts
- CDN integration for fast mobile app asset delivery

REQ-INF-002: Data Model Core Entities

- User: customer accounts with address verification status
- Vendor: partner storefronts with product catalogs
- Product: individual items within vendor catalog
- Order: transactional record with status machine
- Delivery: execution record linking order to drone and pilot
- Drone: fleet asset with status and maintenance history
- Pilot: operator profile with shift history and certifications
- Transaction: financial record of payments and settlements

REQ-INF-003: API Requirements

- RESTful API with JSON responses
- Authentication via JWT bearer tokens
- Rate limiting: 100 requests per minute per user, 1000 per minute per admin
- API documentation maintained in OpenAPI format

REQ-INF-004: Third-Party Integrations

- Paystack for payment processing
- Twilio Africa for SMS OTP and notifications
- Google Maps for address verification and distance calculation
- OpenWeatherMap for weather data feeding dispatch decisions

REQ-INF-005: Security and Compliance

- All data in transit encrypted via TLS 1.3
- Passwords hashed with bcrypt (minimum cost factor 12)
- PCI DSS compliance via Paystack tokenization (no card data stored)
- Nigerian Data Protection Regulation (NDPR) compliance for user data
- Daily automated database backups retained 30 days

4. Phase 2 Functional Requirements

Phase 2 build window: May 29 — June 30, 2026 (4 weeks). Focus: beta testing, drone integration, automation, soft launch readiness.

4.1 Customer App Phase 2 Additions

REQ-CM2-001: Real-Time Drone Tracking

- Live map view during active delivery showing drone location
- Updated every 10 seconds from drone GPS feed
- Estimated arrival countdown timer
- Route visualization from pickup to delivery

REQ-CM2-002: Push Notifications

- Native iOS and Android push notifications via Firebase Cloud Messaging
- Notification triggers: order confirmed, dispatched, drone in flight, 5 minutes to arrival, delivered
- User notification preferences manageable in app settings

REQ-CM2-003: Post-Delivery Rating and Feedback

- Rating prompt appears in app after delivery status changes to delivered
- 5-star rating with optional written feedback
- Feedback routed to operations dashboard for review
- Ratings contribute to public vendor scores

REQ-CM2-004: Saved Favorites and Quick Reorder

- Users can favorite vendors for quick access
- Past orders have Reorder button to duplicate cart
- Saved addresses editable with default designation

REQ-CM2-005: In-App Customer Support Chat

- Support chat widget accessible from any screen
- Chat routed to Support admin role on operator dashboard
- Chat history persists per user
- Automated first response acknowledging inquiry received

4.2 Operator Dashboard Phase 2 Additions

REQ-OD2-001: Automated Dispatch Logic

- System automatically matches incoming orders to available drones based on proximity, battery level, and capacity
- Automated pilot assignment from on-shift roster with load balancing
- Operator can override automated assignments
- Exception alerts when automation fails (no available drones, no on-shift pilots)

REQ-OD2-002: Pilot Performance Analytics

- Per-pilot dashboards showing deliveries completed, on-time rate, customer ratings

- Shift-level performance metrics
- Pilot leaderboards for monthly recognition

REQ-OD2-003: Vendor Partner Portal

- Self-service vendor login to manage their storefront
- Add, edit, remove products from catalog
- Upload product images directly
- View order history and pending fulfillments
- Download transaction reports

REQ-OD2-004: Customer Support Ticket Workflow

- Support tickets generated from in-app chat, email, or manual creation
- Ticket assignment to Support admin role
- Status workflow: open, in progress, resolved, closed
- SLA tracking with alerts for tickets exceeding 4-hour response time

REQ-OD2-005: Financial Reconciliation Dashboard

- Automated settlement tracking from Paystack
- Vendor payout calculations with commission deductions
- Pilot compensation calculations based on completed deliveries
- Monthly financial summary reports exportable to CSV and PDF

4.3 Drone Operations Integration (Phase 2)

REQ-DR-001: Drone API Integration

- Integration with drone manufacturer SDK for fleet of 30 units
- Automated pre-flight diagnostic checks before dispatch
- Battery level real-time feed to operator dashboard
- Weather-based flight cancellation automation using OpenWeatherMap feed

REQ-DR-002: GPS Tracking Data Feed

- Drone location streamed to backend every 10 seconds during active flights
- Location data pushed to customer app for real-time tracking
- Historical flight paths stored for operations analysis

REQ-DR-003: Battery Swap Station Integration

- Battery station status monitoring (available batteries, charging status)
- Pilot coordination alerts when swap is required
- Maintenance scheduling based on battery cycle counts

4.5 AI-Powered Autonomous Logistics Layer (Phase 2)

This section defines the AI logistics intelligence layer that differentiates Skyrunner as a next-generation logistics platform. Technology stack selection will be finalized during Phase 2 Week 7 through structured evaluation across candidate technologies including OpenClaw (open-source agent framework), Google OR-Tools (routing optimization),

AWS Location Service (mapping and geofencing with Nigerian data residency), and commercial drone fleet management platforms. Final selection criteria include security audit results, NDPR compliance, integration complexity, and operational cost.

REQ-AI-001: Autonomous Dispatch Intelligence

- System automatically evaluates all incoming orders against available drone fleet status in real time
- Dispatch algorithm considers: drone location, battery level, cargo weight capacity, pilot availability, delivery urgency, and estimated route duration
- Automated dispatch decisions logged with full reasoning for audit and continuous improvement
- Human operator override available from dashboard with single click
- Exception alerts raised when no optimal match exists (all drones busy, battery constraints, weather restrictions)

REQ-AI-002: Dynamic Flight Path Optimization

- Each dispatched delivery receives an AI-optimized flight path calculated at time of dispatch
- Optimization factors: weather conditions from OpenWeatherMap, NCAA airspace restrictions, battery conservation, delivery time window, return-to-base routing
- Flight paths stored with delivery record for regulatory compliance audit
- Route updates possible mid-flight if conditions change (weather deterioration, airspace alerts)
- Visual route preview available in operator dashboard before dispatch confirmation

REQ-AI-003: Predictive Demand Modeling

- System learns order patterns by time of day, day of week, weather conditions, and resident behavior
- Pre-positioning recommendations generated daily for drone placement at peak-demand zones
- Demand forecasts visible to operations team for shift planning
- Model retraining triggered weekly using prior week's delivery data

REQ-AI-004: Continuous Learning and Performance Improvement

- Every completed delivery feeds performance data back to dispatch and optimization models
- Monthly AI performance reports: dispatch accuracy, route efficiency, prediction vs actual demand
- Operator feedback on AI decisions captured via thumbs-up/thumbs-down in dashboard
- Feedback data incorporated into model refinement cycles

REQ-AI-005: Natural Language Operator Interface

- Operators can query platform status using natural language (e.g., Which drones are available in the next 30 minutes?)
- AI responds with formatted answers and links to relevant dashboard views
- Anomaly explanations: when AI flags exceptions, natural language descriptions provided for operator context
- Integration with Claude API or equivalent LLM for natural language processing

REQ-AI-006: Security and Compliance Guardrails

- AI decision-making restricted to defined action space (dispatch, routing, escalation) — no autonomous access to customer payment data, personal messaging, or external systems
- All AI decisions logged with full context for audit trail
- Human-in-the-loop required for: refund decisions, customer complaint resolution, pilot schedule changes, vendor terminations
- Prompt injection protection via input sanitization and restricted tool access

- Quarterly security audit of AI layer by external cybersecurity consultant
- NDPR compliance maintained through data residency controls on selected AI infrastructure

4.4 Beta Testing Requirements

REQ-BT-001: Closed Beta Program

- Minimum 50 beta users recruited from Banana Island Estate
- Beta feedback form integrated into app
- Weekly beta check-ins conducted by operations team
- Beta users receive 3 free deliveries as compensation

REQ-BT-002: Beta Success Criteria

- 100 successful test deliveries completed
- 95 percent on-time delivery rate maintained
- Zero package loss or theft incidents
- Average customer rating 4.5 stars or higher
- No critical bugs open at end of beta window

5. Technical Architecture

5.1 Technology Stack

Layer	Technology	Rationale
Customer Mobile App	React Native with Expo	Cross-platform iOS and Android, rapid development, Adaobi fluency
Operator Dashboard	Next.js 14 with React and Tailwind CSS	Fast development, excellent DX in Cursor IDE, SEO ready for future public pages
Backend API	Node.js with Express or Next.js API routes	Single-language full stack reduces context switching
Database	PostgreSQL (Neon or Supabase managed)	ACID compliance for financial transactions, mature ecosystem
Authentication	Clerk or Auth.js with JWT	Battle-tested, rapid integration, SMS OTP built-in
Payments	Paystack primary, Flutterwave secondary	Nigerian market standard, strong compliance, local support
SMS / Notifications	Twilio Africa, Firebase Cloud Messaging	Reliable delivery to Nigerian carriers, native mobile push
Hosting	Vercel (frontend), AWS (backend services)	Vercel for speed of deployment, AWS for flexibility and regional options
Monitoring	Sentry for errors, Vercel Analytics, Datadog (Phase 2)	Early error visibility critical for rapid iteration
AI Logistics Layer (Phase 2)	Candidates under evaluation: OpenClaw, Google OR-Tools, AWS Location Service, commercial drone fleet platforms	Structured evaluation Phase 2 Week 7. Selection criteria: security audit, NDPR compliance, integration complexity, operational cost
LLM Integration	Claude API (Anthropic) primary, OpenAI GPT secondary	Natural language operator interface, anomaly explanations, decision reasoning

5.2 System Architecture Overview

Skyrunner follows a modular 4-tier architecture with the AI logistics layer as a dedicated intelligence tier sitting between the application and data layers:

- Presentation Layer: Customer React Native app, Operator Next.js dashboard, Vendor Portal (Phase 2)

- Application Layer: Node.js API services, business logic, dispatch orchestration, payment processing
- AI Intelligence Layer (Phase 2): Autonomous dispatch engine, flight path optimizer, predictive demand models, LLM-powered operator interface
- Data Layer: PostgreSQL primary database, S3 object storage, Redis cache (Phase 2), vector database for AI memory and learning (Phase 2)

External service integrations sit alongside the application and AI layers: Paystack payment processor, Twilio SMS gateway, Google Maps API, OpenWeatherMap, drone manufacturer SDK (Phase 2), LLM APIs (Claude or OpenAI) for natural language processing, optional OpenClaw or Google OR-Tools for logistics decision engines.

6. Development Timeline and Milestones

6.1 Phase 1 Weekly Milestones (6 weeks)

Week	Focus	Deliverable
1	Foundation: repo setup, database schema, authentication, NCAA application filed	User registration and login functional; NCAA package submitted
2	Customer app: address verification, vendor catalog browsing, cart functionality	Customer can browse vendors and add items to cart
3	Payment integration, order placement, notifications foundation	End-to-end order flow working with Paystack sandbox
4	Operator dashboard: order queue, manual dispatch, fleet status view, pilot shift calendar	Operator can accept orders and assign drones + pilots
5	Custom delivery requests, order history, revenue dashboard, Paystack production	Production payment processing live; complete Phase 1 feature set
6	Internal QA, bug fixes, closed beta onboarding (5 to 10 testers), vendor partner onboarding	Phase 1 platform stable; 3 vendors integrated; 5 to 10 beta users active

6.2 Phase 2 Weekly Milestones (4 weeks)

Week	Focus	Deliverable
7	Real-time tracking, push notifications, drone API integration begins, AI logistics technology evaluation and selection	Customers see live drone locations; AI framework selected with documented evaluation
8	Automated dispatch logic with AI, vendor portal, ratings and feedback, drone fleet testing with hardware, flight path optimization live	AI-powered dispatch reduces manual workload by 70 percent; 30 drones operational
9	Support chat, customer service workflow, financial reconciliation, pilot analytics, predictive demand modeling baseline	Full operations workflow live; 50+ beta users actively using platform; AI demand predictions producing daily recommendations
10	Soft launch preparation, marketing campaign execution, load testing, final QA	SOFT LAUNCH on Banana Island Estate June 30, 2026

7. Out of Scope for MVP

The following capabilities are explicitly deferred beyond Phase 1 and Phase 2 MVP launch:

- Expansion to Ikoyi and Lekki delivery zones (planned Q3 2026)
- Premium subscription tier with unlimited deliveries (planned Month 6 post-launch)
- B2B contracted delivery for corporate clients (planned Month 9 post-launch)
- Advanced AI capabilities beyond Phase 2 baseline: reinforcement learning from delivery outcomes, multi-drone coordinated missions, autonomous vendor onboarding (Phase 3)
- Multi-language support beyond English (Phase 3)
- Drone maintenance scheduling automation with IoT sensors (Phase 3)
- International expansion (Ghana, Kenya, South Africa) (2027 roadmap)
- Vendor onboarding self-service (Phase 2 includes portal; full automation in Phase 3)

8. Dependencies and Assumptions

8.1 External Dependencies

- Nigerian Civil Aviation Authority (NCAA) approval of commercial UAS operator application by end of Week 8
- Paystack merchant account approval by end of Week 3
- Attarhi Limited Banana Island Estate resident registry access for address verification
- Drone hardware supplier delivery timeline meeting Phase 2 Week 8 fleet availability
- Insurance underwriter issuance of fleet, pilot, and cargo liability coverage by Week 9

8.2 Assumptions

- Adaobi has uninterrupted focus for platform build during Phase 1 with minimum 30 hours per week
- Claude Code Max and Cursor Pro tools accelerate development by 3x over solo coding baseline
- Attarhi Limited releases Phase 1 capital (\$11,200) within 7 days of Business Plan approval
- Attarhi Limited releases Phase 2 capital (\$97,800) by Week 6 end for drone hardware procurement
- Banana Island Estate Board approves airspace access for designated delivery corridors by Week 6

8.3 Critical Path Items

The following items sit on the critical path and require proactive monitoring:

- NCAA regulatory approval (Weeks 1 through 8) — single biggest timeline risk
- Drone hardware procurement and delivery (Weeks 6 through 8) — long lead time international shipment
- Attarhi Estate airspace access approval (Weeks 4 through 6) — required before live test flights
- Insurance coverage finalization (Weeks 7 through 9) — cannot launch without coverage

9. Appendix

9.1 Revision History

Version	Date	Author	Changes
1.0	April 16, 2026	Adaobi Onyekaba	Initial PRD for Phase 1 and Phase 2 MVP scope

9.2 Glossary

- MVP — Minimum Viable Product
- NCAA — Nigerian Civil Aviation Authority
- UAS — Unmanned Aircraft System (drone)
- NDPR — Nigerian Data Protection Regulation
- NGN — Nigerian Naira
- UHNW — Ultra High Net Worth
- SLA — Service Level Agreement
- DSP — Demand Side Platform
- PRD — Product Requirements Document
- LLM — Large Language Model (e.g., Claude, GPT)
- OpenClaw — Open-source autonomous AI agent framework (Phase 2 evaluation candidate)
- OR-Tools — Google's open-source operations research library for routing optimization

9.3 Approval

This PRD is approved for Phase 1 and Phase 2 execution upon sign-off by both parties in the accompanying Business Plan document.

Chukwujekwu — God has the final say.